

Jakub Jan Duchniewicz

SENIOR EMBEDDED AND SOFTWARE ENGINEER · TEAM LEADER

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Summary

I am a Senior Embedded Engineer and (sometimes) a team leader at Tietoevry, specializing in developing L1 features for 5G NR on custom ASICs and designing complex systems. With an MSc in Embedded Systems and a minor in Innovation and Entrepreneurship, I bring a strong technical background and innovative mindset to my work.

My expertise spans embedded systems, FPGA, Linux, audio/video processing, game engine development, and machine learning. Proficient in C, C++, and Rust, I focus on embedded machine learning, security, and reverse engineering.

As a co-founder of Sticky Piston Studios, I innovate at the intersection of technology and creativity. I've presented at conferences such as the Embedded Open Source Summit, FOSDEM or State of Open Con, discussing AI-powered wearable health monitors and hardware acceleration using FPGAs, showcasing my frequent engagement with the tech community.

Mentoring is a passion, and I regularly share my knowledge through blog posts on my personal website. Additionally, I am an avid game jammer, consistently winning prizes for my creative and technical contributions.

Experience

Tietoevry

Remote - Wrocław, Poland

SENIOR EMBEDDED AND SOFTWARE ENGINEER

Dec. 2021 - Present

- Leading, supporting and coaching a team of engineers.
- Implementing 5G features on a custom ASIC modem.
- Debugging RTL and C code in Simvision.
- Converting 3GPP specification algorithms to C code.
- Development and maintenance of ORAN compliant FlexRAN fork.
- Implementation of novel ORAN features in translation module between IAPI and FAPI.
- Architecting, planning and implementing L2 ORAN compliant application in C++.
- Porting FlexRAN project from icc to icx compiler.
- Fuzzing of L1-L2 interface in FlexRAN with libfuzzer.
- Debugging and fixing issues found in DPDK drivers.

United Micro

Remote

SENIOR EMBEDDED ENGINEER

Jul. 2024 - Present

- Support in porting code from earlier architecture to 5G eRedCap solution.
- Writing L1 synchronization layer and firmware code.
- Various memory and speed of operation optimization and fixes.

Sequans

Remote

SENIOR EMBEDDED ENGINEER

Nov. 2023 - Jun. 2024

- Designing and delivering a bare-metal modem driver.
- Developing the Blind Decoding Algorithm for PDCCH 5G PHY channel.
- Validating custom ASIC RTL in conjunction with my driver.
- Discovering bugs, debugging and providing fixes for various parts of a complex real-time system.

Intel Corporation

Remote - Wrocław, Poland

SENIOR EMBEDDED AND SOFTWARE ENGINEER

Jun. 2022 - Oct. 2023

- Leading, supporting, and coaching a team of developers and managing projects.
- Recruiting engineers of all levels of expertise.
- Co-leading the ORAN M-Plane project.
- Developing the FlexRAN 5G solution.
- Implementing L1 related features.
- Implementing L2 related features.
- Conducting fuzz testing using libfuzzer and radamsa.
- Performing low-level optimization of L1 functionality.

Sticky Piston Studios

[Remote](#)

SOFTWARE ENGINEER

Dec. 2020 - Present

- Design and creation of City Event Map - alongside with deployment in the Region of Vilnius on Lithuania.
- Developing web applications, including frontend, backend, databases, and dev ops.
- Designing UX and UI for digital products.
- Attending and participating in game jams and hackathons.

Google Summer of Code, beagleboard.org

[Remote](#)

EMBEDDED SOFTWARE ENGINEER

May. 2021 - Aug. 2021

- Using OpenGL ES 2.0 and EGL for GPGPU computation accelerations on BeagleBone Black with SGX 5xx GPU's.
- Wrote library in C which makes these computations easier.
- Implemented most popular computations (scalar operators, array operations, 2D convolution).
- Wrote both single-shot and chain API (for combining computations).
- Benchmarked the library on various data sizes.
- Documented the project on a blog.

Samsung Electronics

[Warsaw, Poland - Suwon, South Korea](#)

JUNIOR SOFTWARE ENGINEER

Feb. 2018 - Mar. 2020

- Improved Tizen Operating System Broadcast middleware (C++/C/Arm assembly).
- Managed performance in an embedded system, through code restructurization and thread-wise code improvement.
- Improved performance of code in major broadcasting pipelining module by aware refactorization and smart usage of C++17 STL and Boost library.
- Improved overall health of code, by analysis of coredumps and various system logs. When deemed necessary provided other teams with ready solution.
- Developed from scratch a multithreaded C and C# compliant middleware process, with custom threaded work queue implementation.
- Worked with version control tools (git, p4) and in an agile environment (scrum).
- Developed bash and python scripts aiding development and analysis of problems.
- Developed python utility program for automation of defect analysis.
- Traveled to South Korea to aid with defects management during commercialization and to offer quick support.
- Worked in a diverse multi-cultural environment (directly with a group of 30 people), giving helping hand those who needed it.

Samsung Electronics

[Warsaw, Poland](#)

INTERN

Dec. 2017 - Feb. 2018

- Supported in project management and knowledge transfer from other R&D Institute.
- Based on good performance and analytical skills got an offer for full-time work.

BoSport/Beskid Ski Arena

[Chałupy/Szczyrk, Poland](#)

KITE/WIND/SKI INSTRUCTOR

Jun. 2014 - Sep. 2017

- Worked with all kinds of people, teaching them and maintaining good relations throughout the years following.
- Taught in various changing conditions, always taking responsibility for life of others.
- Taught in both Polish and English in groups up to 10 people.

Education

EIT Digital Master School

[Europe](#)

M.SC. IN EMBEDDED SYSTEMS

Sept. 2020 - Nov. 2022

- Holder of EIT Excellence Scholarship
- Winner of Digital Health Summer School with "Medpipe" - personalized tracker for recovery in endoprosthetic surgeries.

KTH Royal Institute of Technology

[Stockholm, Sweden](#)

M.SC. IN EMBEDDED SYSTEMS

Sept. 2021 - Nov. 2022

FPGA ACCELERATED PACKET CAPTURE WITH EBPF.

- Implemented packet capturing offload from Linux network stack to FPGA.
- Implemented Avalon MM host and target with bursting and stalling capabilities.
- Patched stmmac Ethernet driver to accelerate ingress packets on FPGA.
- Wrote userspace .pcap capturing and testing programs.
- Tested the solution using ModelSIM and SignalTap simulators.
- Achieved performance gains while maintaining power consumption.

University of Turku

[Turku, Finland](#)

M.SC. IN EMBEDDED SYSTEMS

Sept. 2020 - Aug. 2021

Warsaw University Of Technology

B.Sc. IN COMPUTER SCIENCE AND NETWORKS

FPGA BASED HARDWARE ACCELERATOR FOR MUSICAL SYNTHESIS FOR LINUX SYSTEM.

Warsaw, Poland

Oct. 2016 - Aug. 2020

- Nomination to IEEE Polish Diploma Contest
- Implemented DDS sythesis, filtering and sample accumulation in Verilog in a streamlined fashion.
- Implemented polyphony and various waveform shapes.
- Synchronized the programmable logic with the MCU.
- Deployed the solution in the De0 Nano SoC FPGA.
- Implemented Linux kernel drivers for communication with the FPGA and ALSA subsystem.
- Implemented the ALSA soundcard driver.
- Wrote userspace application for communicating MIDI commands to the FPGA via a kernel driver.
- Tested the solution to ensure smooth and high-fidelity sound.

Extracurricular Activity

BIBoP

PROGRAMMING AND ELECTRONICS LEAD

Warsaw - Poland

Mar. 2021 - Jun. 2021

- Wrote project architecture and data collection from sensors.
- Created a ML model for Blood Pressure predictions from PPG.
- Deployed the model on AWS Lambda with custom hooks.
- Implemented networking over MQTT protocol.
- Soldered and assembled the prototype.
- Designed and soldered analog Galvanometer.

Envidrawer

PROGRAMMING AND ELECTRONICS LEAD

Warsaw - Poland

Sept. 2020 - Jan. 2021

- Designed a sustainable, automated plant growing solution working in varying climate conditions.
- Programmed a sensor-monitoring system and a visualisation web-based service.
- Designed and implemented an analog control circuit as well as 12 V powering circuit.
- Nurtured friendly atmosphere and healthy work-life balance during the project with 2 other members.

BEYOND 2030 Challenge

MENTOR

Turku - Finland

Oct. 2020 - Nov. 2021

- Guided a group of 2 high-school students in their environmentally sustainable project - Smart bee-hive.
- Provided support in technical and project management matters.
- Gave support where needed, pulling the students instead of pushing.

PolyEngine

DEVELOPER

Warsaw - Poland

Oct. 2017 - Present

- Reimplemented Entity Component System systems.
- Refactored code and cleaned architecture of engine.
- Developed various necessary features during gamejams (both gameplay and engine-wise).
- Developed 3 games during Global Game Jam 18/19 and Slavic Game Jam 18.
- Developed efficient data structures for a game engine: string and queue with STL compliance.

KNTG Polygon

VICE PRESIDENT AND EVENT ORGANISER

Warsaw - Poland

Oct. 2017 - Jun. 2019

- Organised weekly meetings for student game development group Polygon.
- Gave two talks on Modern C++ usage and easily overlooked nuances.
- Planned and realised two 4-meeting editions of Game Dev Fest: invited lecturers and managed sponsorship of both events.
- Managed sponsorship funds and promotion of both events.
- Helped organise game jams (Polyjam 2018/2019).

Presentations

Digital Futures

PRESENTER FOR *Open Source in the Development of 5G and LTE Networks*

Warsaw, Poland

May. 2024

- Discussed open source solutions for modern telecommunications networks, emphasizing the benefits and challenges of integrating these technologies.

Embedded Open Source Summit 2024

Seattle, WA, USA

PRESENTER FOR *ZLED Frame or how to illuminate your artwork*.

Apr. 2024

- Delivered a detailed session on ease of using Zephyr OS for your own projects with a 3D printed Digital Art frame as an example.

FOSDEM 2024

Brussels, Belgium

PRESENTER FOR *Open Source in the Development of 5G and LTE Networks*

Feb. 2024

- Delivered a talk on the role of open source in 5G and LTE network development, highlighting the importance of an open ecosystem in telecommunications.

Hackaday Supercon 2023

Pasadena, CA, USA

PRESENTER FOR *Porting an AI Powered Wearable Health Monitor to Zephyr on Open Hardware*

Nov. 2023

- Not only gave a talk but also explored creative uses of Vectorscope badge and other innovative hardware hacks.

Digital Futures

Kosice, Slovakia

PRESENTER FOR *Telco Trends and 5G Advancements*

Nov. 2023

- Shared insights on the latest developments in 5G and their impact on daily life, showcasing Tietoevry's innovative approach.
- Provided an in-depth perspective on Telco L1, enhancing the technical discussion.

Embedded Open Source Summit 2023

Prague, Czech Republic

PRESENTER FOR *Porting an AI Powered Wearable Health Monitor to Zephyr on Open Hardware*

June 2023

- Discussed challenges and solutions in porting AI-powered health monitors to Zephyr OS, showcasing practical insights and technical tips.

State of Open Con 2023

London, United Kingdom

PRESENTER FOR *Hardware Acceleration Using FPGAs in Embedded Linux*

March 2023

- Presented a custom Linux distribution for FPGA-based hardware acceleration, focusing on performance metrics and optimization techniques.

Skills

Programming

C++{98,11,14,17,20}, C{89,99,11,17}, C#, Rust, ARM/x86 assembly, AVX2/512, CEVA DSP, Python, Bash, Verilog, SystemVerilog, VHDL, OpenGL ES, Make, CMake, LaTeX

Technical

FPGA, Electronics, Microcontrollers, Bootloaders, SoC, ASIC, RTOS, Linux Kernel, DSP, ROS, CUDA, PyTorch, Deep Learning, DSP, Hardware Accelerators, GPGPU, Network Stack, 5G L1/L2, DPDK, Fuzzing, IoT, Compilers, Game Engine Architectures

Languages

Polish, English, German, Spanish

Honors & Awards

Dec. 2023 **Award in Data for City Hackathon 2023**, Urząd m.st. Warszawy

Warsaw, Poland

City Event Map is an innovative web and mobile application designed to uplift the quality of life for city residents. It consolidates information about local cultural events, roadworks, service interruptions, and more, simplifying information access, improving events visibility, and effectively eliminating citizens' confusion. Beyond its role as a central information hub, City Event Map acts as a vital link between municipal authorities and residents, fostering transparent and efficient communication within the community.

Nov. 2022 **Winner - Game Development Challenge**, HackYeah

Kraków, Poland

Won the Game Development category during the largest stationary Hackathon in Europe - HackYeah. The winning game can be found here: The Deluge.

2021 **Winner - Best Entrepreneurial Team**, EIT Digital Summer Health School 2021

Talinn, Estonia

"Medpipe" - an application that will help people managing their treatment timeline and supporting materials (diet, exercises, lifestyle) before and after surgery.

2019 **Honorable mention**, Warsaw Film School Game Jam VI

Warsaw, Poland

"Eternal Feud" - Low-poly, competitive co-op, made in a duo team.

2018 **Honorable mention**, Static Code Analysis Competition - Samsung Electronics

Global

2017 **2nd Place Overall, 2nd Place in Innovation**, Polyjam, Global Game Jam 2017

Warsaw, Poland

"Ommm" - Controlling a monk with the power of your voice.